

Project Record

Rationalizing Tariff Structures in the Indian Downstream Aluminium Industry Analytical work, reportage and visualisation

At a glance. An evidence-based policy submission arguing for rationalisation of the basic customs duty on primary aluminium (HS 7601). The work comprised: an effective-rate-of-protection analysis across thirteen product lines and six trading partners; an industrial-statistics analysis built from 3,705 observations of the Annual Survey of Industries; a public-finance analysis of the tariff-versus-GST trade-off; and the production of a 45-page report with 21 figures, supported by an automated verification harness and three technical companion documents.

1 Objective and scope

The study set out to answer four questions: whether India's present tariff structure places downstream aluminium products at a measurable disadvantage; what that disadvantage costs in value addition and employment; whether the fiscal objection to correcting it holds; and what policy change follows. Scope was fixed at HS 7601–7616, the chapter-76 lines spanning unwrought metal through fabricated articles.

The central analytical problem was that the tariff evidence and the industrial evidence had to describe *the same industry*. Resolving that — deriving the industrial classification from the tariff schedule rather than from convention — was the methodological core of the project.

2 Data sources

Source	Coverage	Used for
Annual Survey of Industries, All-India summary	3,705 observations; 15 indicators, 19 NIC classes, 13 years (2011–12 to 2023–24)	All industrial statistics: value added, employment, wages, workers, contract labour, net profit
MOSPI Supply–Use Tables	26 supply and use files, 2011–12 to 2023–24	Effective protection estimates; input-cost shares
Tariff schedules	256 tariff lines, HS 7601–7616, six partners	MFN and preferential rates; the ERP basket
Effective corporate tax rates	By year, 2011–12 to 2023–24	Corporate-tax base estimation
Ministry of Mines, OECD, GTRI, CEA, Union Budget	Published	Capacity, demand, trade composition, public capital expenditure

3 Analytical work

3.1 Segmentation: deriving the classification from the tariff schedule

The prior draft partitioned the ASI on convention. The work replaced this with a rule derived from Corden's definition of an *activity* — a stage of conversion from traded input to traded output — so that a NIC class enters the downstream aggregate if and only if its principal output falls within the thirteen ERP lines HS 7604–7616. This produced upstream = NIC 2420 and downstream = NIC 2432, 2511, 2599, and required excluding four classes previously included whose outputs fall in HS Chapters 85 and 88.

The rule was cross-checked against the OECD's aluminium value-chain segmentation and against the material-flow literature, and the divergence from the latter was tested rather than argued: reassigning the semi-fabrication class moves the headline result from 3.14 to 2.77 and changes no conclusion.

3.2 Series construction

Thirteen annual series were built for each segment and each constituent class: gross value added, output, inputs, wages, persons engaged, total workers, contract workers, net profit, and estimated corporate tax. From these, nine derived measures: value added per person engaged, labour-intensity ratio, wage share, contract share, input-cost share, employment yield per unit of tax, index series, three-year endpoint growth, and year-on-year log volatility.

3.3 Principal findings

Finding	Result	Robustness
Labour intensity: employment sustained per unit of value added	3.14 × downstream; range 3.08–6.18, median 3.93	Holds in 12 of 12 years; exceeds the Ministry of Mines' 2.5× benchmark in all
Labour share of value added	38.4% vs 17.6%	Series never cross in 12 years; uses only monetary variables
Employment per unit of estimated corporate tax	151 vs 39 persons per crore	Ordering holds in 12 of 12 years
Structural products (NIC 2511, HS 7610)	+15.0% nominal over 11 years; workforce flat	The single most disprotected ERP line
Volatility of value added	Upstream 3.6 × downstream	sd of year-on-year log growth
Input-cost amplification	4.91 ×	Derived from a measured input share of 83.1%

3.4 Analytical results established rather than asserted

Three results were derived formally rather than argued or simulated, and these are the most durable output of the project.

1. **Apportionment invariance.** The objection that the NIC classes are not exclusively aluminium was closed by proof rather than sensitivity runs. An apportioned segment productivity is necessarily a weighted average of its class productivities, hence bounded by their range for *any* positive apportionment vector; the upstream segment being a single class, its coefficient cancels. The labour-intensity ratio therefore lies in [3.01, 3.38] whatever apportionment is assumed. Verified against 200,000 random draws.
2. **The per-tonne decomposition.** The employment-per-tonne ratio factors exactly into the per-rupee ratio times value added per tonne ($3.14 \times 1.81 = 5.7$). This reconciled the report's own measure with the much larger ratios circulating in industry discussion, and showed those to arise from comparing mismatched employment universes.
3. **The transfer-to-revenue identity.** In the tariff-versus-GST analysis, the ratio of surplus transferred to revenue collected reduces to $a/c = \phi(1 - \mu)/[\mu(1 - f)]$ — import penetration, pass-through and the duty-free share. Price, exchange rate and both elasticities cancel. The result is therefore robust to the parameters least well known.

3.5 Public-finance analysis

A separate analysis addressed the fiscal objection to rationalisation: partial equilibrium welfare accounting (producer surplus, revenue, and both deadweight triangles, using supply and demand elasticities from the metals literature), the revenue-replacement arithmetic against the GST base, and an induced-expansion offset derived from the amplification factor. Headline results: **INR 17.5** transferred per rupee collected, **54 paise** of national income destroyed per rupee raised, and a revenue requirement replaceable by **0.18 of a percentage point** of GST.

4 Academic grounding

The argument was grounded in primary sources rather than secondary summary, with 27 references in the final bibliography. The load-bearing citations:

Source	Role in the argument
Corden (1966)	Defines the activity as the unit of effective-protection analysis; supplies the segmentation rule
Balassa (1965, 1968)	Precedent for negative effective protection; tariff escalation
Pathania & Bhattacharjea (2020)	Modified Corden estimator; Indian inverted duty structures
Diamond & Mirrlees (1971)	Production efficiency: intermediate goods should not be taxed
Bhagwati & Ramaswami (1963)	Target the distortion at source; trade instruments are second-best
Emran & Stiglitz (2005); Keen (2008)	The informality objection to VAT-for-tariff substitution, and its limits
Baunsgaard & Keen (2010)	Empirical evidence on revenue recovery after liberalisation
Bohlin & Widell (2006); Miller & Blair (2009)	Technology assumptions; factor-content invariance
Cullen & Allwood (2013); Bertram et al. (2009)	Material-flow taxonomy of the aluminium system
Chaurey (2015); Goldberg et al. (2010)	Contract labour; imported intermediate inputs and product growth
OECD (2019); Viner (1950)	Value-chain segmentation; trade diversion

5 Visualisation

Twenty-one figures across the report, seven of them rebuilt for the industrial statistics section. All are vector charts generated directly from the computed series, with the plotting coordinates emitted by the analysis code so that a figure cannot drift from its data.

5.1 Design constraints and how they were met

1. **Monochrome legibility.** The report is a policy submission that will be printed and photocopied. Every series is distinguished by three redundant cues — line style, marker shape and marker fill — so that no series depends on colour or grey level. Bar charts pair a hatched light fill against a solid dark one.
2. **No text merging with the plot.** Every printed value sits on an opaque white chip, so a plot line or gridline passing beneath cannot obscure it. Where two series run close or cross, per-point labelling was replaced by direct annotation of endpoints and turning points.
3. **Legends outside the plotting area,** never overlying data.
4. **Values on the chart.** No number in a figure has to be read off an axis.
5. **Captions carry a source line only.** All construction, derivation and domain detail was moved to the appendix, so that the narrative reads without interruption.

5.2 Editorial decisions on the figure set

The section was reduced from eight figures to seven single-panel figures, cutting Chapter 4 from roughly seventeen pages to nine. Three charts were removed on the principle that a figure earns its place only if it advances the argument: a value-added-per-person

chart whose ratio was the following figure; a persons-engaged index panel superseded by the labour-intensity chart; and a corporate-tax *levels* panel, on which the comparison runs against the report's own case — retained as a sentence of text so the concession is made, but not given a chart.

6 Quality assurance

6.1 The verification harness

An automated checker recomputes every series from source, parses every plotting coordinate out of the document, and tests whether each plotted number is a correct rounding of the computed value *at the precision printed*. It also re-checks the statistics quoted in prose. Current status: **all checks pass across 158 plotted points and 30 quoted statistics**.

6.2 Errors found and corrected

The harness caught four errors that had survived several drafts: three plotted points rounded twice (136.47 printed as 137, 214.47 as 215, 139.47 as 140), and an amplification factor computed from a rounded input share and printed as 4.92 instead of 4.91. A separate manual check found a caption claiming NIC 2511 spent eight of eleven years below its base level, where the correct count is seven. None changed a conclusion; none would have been found by reading.

6.3 Analytical corrections made during review

Two substantive errors were caught and corrected in the analysis itself: a robustness claim that conflated leave-one-out with reassignment of a class (3.13 stated where 2.77 was correct), and a transfer-to-revenue ratio computed on downstream fabrication throughput where the correct base is all domestic primary metal sold domestically (4.7:1 stated where 17.5:1 was correct).

6.4 Standards of claim

Causal language was audited and withdrawn where it had not been earned. The report now states explicitly that its evidence is association rather than an identified causal effect, names the absent counterfactual and the uncontrolled construction cycle, and removes a timing claim that twelve annual observations cannot support. Known weak points — an assumed parameter, an industry-estimated tonnage, a tax proxy, a frame that excludes unincorporated units — are stated in the documents rather than left to be discovered.

7 Deliverables

Document	Extent	Purpose
Main report	45 pp, 21 figures	The policy submission
Author's notes	22 pp	Every calculation behind every figure, with worked arithmetic; variable and coefficient glossary; the arguments and the figure carrying each
Tariff-versus-GST note	8 pp	Public-finance analysis; declared baseline, welfare accounting, sensitivity
Decision record	2 pp	Reviewer comments: what was adopted, what was not, and why
<i>Analysis and build code</i>		
Series computation	—	Builds all series; emits plotting coordinates
Verification harness	—	Independent re-derivation and rounding audit
Report assembly	—	Deterministic build with cross-reference and bibliography injection
Public-finance arithmetic	—	Declared baseline, results, sensitivity

8 Summary lines, for a CV or capability statement

- Built the industrial-statistics evidence base for a policy submission on aluminium tariff reform, constructing thirteen-year series for nine measures from 3,705 ASI observations and reconciling the industrial classification to the tariff schedule so that the statistical and trade evidence described the same industry.
- Established the central finding — that value added in downstream fabrication sustains 3.1 times the employment of value added in primary metal — and demonstrated it held in every year of the record, corroborating an official estimate from an independent source.
- Closed the principal methodological objection analytically rather than by simulation, proving the headline ratio bounded within [3.01, 3.38] for any apportionment of the underlying classes, and verifying the bound against 200,000 draws.
- Conducted a public-finance analysis of the tariff-versus-GST trade-off, deriving a transfer-to-revenue identity independent of price, exchange rate and elasticity, and quantifying the revenue at issue as replaceable by 0.18 of a percentage point of GST.
- Designed and produced 21 monochrome-legible vector figures under print constraints, with plotting coordinates generated directly from the analysis so that figures could not drift from their data.
- Built an automated verification harness that re-derives every plotted point and quoted statistic from source; it caught four numerical errors that had survived several drafts.
- Grounded the argument in primary academic sources across trade theory, optimal taxation and industrial ecology, and audited the report's causal language, withdrawing claims the data could not support.